Dissecting Treatment Response: The BR-ER-TV Framework and Gompertz Trajectory Modeling in N-of-1 Trial Design

A Clinician's Guide to the Hendrickson et al. (2020) Simulation, with Worked Examples from the Raskind et al. (2013) Prazosin Trial

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1 Why This Document Exists

When a patient improves on a medication, the natural clinical question is straightforward: did the drug work? The answer, however, is rarely simple. Clinical improvement is a composite of at least three distinct processes operating simultaneously, and standard parallel-group randomized controlled trials (PG-RCTs) are poorly equipped to disentangle them at the individual level.

Hendrickson et al. (2020) proposed an aggregated N-of-1 trial design that addresses this limitation directly. Their simulation study, published in *Frontiers in Digital Health*, compared the statistical power of four trial architectures for validating a predictive biomarker – standing systolic blood pressure – against treatment-specific response to the PTSD pharmacotherapeutic prazosin. The simulation parameters were grounded in data from the Raskind et al. (2013) parallel-group RCT of prazosin in active-duty soldiers with combat-related PTSD (*Am J Psychiatry* 170:1003–1010).

This document serves two purposes. First, it provides an accessible exposition of the three-factor decomposition (BR, ER, TV) that underlies the Hendrickson simulation, with concrete clinical examples drawn from the prazosin trial. Second, it reconstructs the Gompertz trajectory parameters from the published Raskind data, allowing the reader to see exactly how abstract simulation inputs map to observable clinical outcomes. The intended audience is clinicians with moderate statistical training who wish to understand the Hendrickson paper at a level deeper than the abstract but without requiring facility in simulation programming.

2 The Clinical Problem: What Does 'Response' Actually Mean?

2.1 A Familiar Scenario

Consider a soldier returning from deployment in Iraq with severe combat-related PTSD. His Clinician-Administered PTSD Scale (CAPS) total score is 82 at baseline – well above the 50-point threshold used by Raskind et al. (2013) for trial eligibility. He is enrolled in a 15-week double-blind RCT of prazosin, an alpha-1 adrenoreceptor antagonist originally developed for hypertension, now repurposed for PTSD nightmares. His standing systolic blood pressure at baseline is 130 mmHg.

After 15 weeks on prazosin, his CAPS score drops to 55 – a 27-point improvement. Is this the drug working?

The honest answer is: partly. His improvement is an inseparable mixture of at least three processes:

1. **The drug itself** reduced his hyperarousal and nightmares by blocking alpha-1 adrenergic receptors in the CNS (the *biological response*).
2. **His expectation of benefit** – the knowledge that he was enrolled in a trial and might be receiving active medication – produced neurobiologically real symptom relief through dopaminergic and anti-inflammatory pathways (the *expectancy-related response*).
3. **The passage of time** – the structure of regular clinical contact, regression to the mean, and natural fluctuation in PTSD severity – contributed its own trajectory of change (the *time-varying response*).

A standard PG-RCT can estimate the *average* drug effect across the group (prazosin minus placebo), but it cannot tell this individual soldier how much of his 27-point improvement was pharmacological and how

much was non-specific. More critically for precision medicine, it cannot determine whether his baseline blood pressure of 130 mmHg *predicted* his drug-specific response.

2.2 Why the Standard RCT Falls Short

The Raskind et al. (2013) trial enrolled 67 soldiers (32 prazosin, 35 placebo) and found a statistically significant 11.3-point advantage for prazosin on the CAPS total score ($p = 0.02$). That group-level finding is valuable but limited in two important ways.

First, participants randomized to placebo provide no information about what their drug-specific response would have been. Second, participants randomized to prazosin who improve cannot be distinguished from participants who would have improved equally on placebo. As Hendrickson et al. (2020) note, this makes PG-RCTs 'poorly optimized for quantifying the relationship of a biomarker measured at baseline to a treatment-specific response.'

The key insight of the N-of-1 approach is that each participant serves as his or her own control, cycling through periods on active drug and placebo. This within-person contrast isolates the biological response from the non-specific components far more efficiently than between-group comparisons.

3 The Three-Factor Decomposition

3.1 Overview

The Hendrickson simulation models clinical symptom change over time as the linear sum of three factors:

$$Y_{it} = \mathrm{BL}_i + \mathrm{BR}_{it} + \mathrm{ER}_{it} + \mathrm{TV}_{it} + \varepsilon_{it}$$

where i indexes participants, t indexes timepoints, BL_i is the baseline symptom level, and ε_{it} is measurement noise. Each of the three response factors captures a distinct causal mechanism.

3.2 Biological Response (BR)

Definition. BR is the direct pharmacological effect of the drug on the patient's physiology. It depends on drug exposure and individual pharmacokinetics, and is independent of the patient's beliefs or expectations.

Clinical example. Prazosin is an alpha-1 adrenoreceptor antagonist that reduces norepinephrine effects at CNS alpha-1 adrenoreceptors (Raskind et al. 2013). Enhanced CNS adrenergic activity contributes to PTSD pathophysiology, particularly the hyperarousal symptoms assessed by the CAPS. When a soldier takes prazosin, the drug accumulates over the 6-week titration period, blocking these receptors with increasing effectiveness. His hyperarousal symptoms decline – not because he believes they will, but because the receptor is physically occupied.

Critically, Raskind et al. (2013) note that prazosin has a 'short duration of action (6 to 10 hours),' necessitating twice-daily dosing. This means BR is pharmacokinetically tied to recent drug exposure. When prazosin is withdrawn, the drug washes out within days and BR returns to zero. This on-off property is precisely what makes crossover and N-of-1 designs informative: the contrast between on-drug and off-drug periods within the same person isolates BR from everything else.

What BR looks like over time. BR does not increase linearly. Each additional day of drug exposure produces diminishing returns as receptors approach saturation. The trajectory follows a saturating curve: rapid initial improvement, then a gradual leveling off toward a biological ceiling. In the Raskind data, the prazosin-minus-placebo difference grew quickly through week 7 and then began to plateau, consistent with this pattern.

3.3 Expectancy-Related Response (ER)

Definition. ER is the component of improvement attributable to the patient's belief or expectation that he or she is receiving effective treatment. This is the placebo effect in its neurobiological sense – not a mere artifact, but a measurable physiological response driven by expectation.

Clinical example. When our soldier is told he will receive either prazosin or placebo, the mere act of enrollment and clinical attention activates hope and expectation. Research in placebo neuroscience has demonstrated that expectation of benefit can activate endogenous opioid and dopamine systems, reduce inflammatory markers, and produce measurable symptom relief. In the Raskind trial, placebo participants improved by 13.8 CAPS points over 15 weeks – a substantial change that cannot be attributed to any pharmacological mechanism.

Why ER depends on trial phase. The magnitude of ER varies with the participant's awareness of treatment assignment:

Trial phase	Expectancy	Reasoning
Open-label (on active drug)	1.0	Full knowledge of active treatment
Blinded (drug vs. placebo)	0.5	Uncertain whether receiving drug
Known placebo or washout	~0	No expectation of drug benefit

This is why the Hendrickson N-of-1 design (their Design 4) begins with an 8-week open-label prazosin phase. During this period, both BR and ER are fully engaged, producing maximum clinical improvement and allowing acutely symptomatic patients to stabilize before entering the blinded phases. The subsequent blinded discontinuation and crossover blocks then manipulate drug exposure while holding expectancy at the intermediate level (0.5), enabling the analysis to separate BR from ER.

Why ER must be modeled separately. If ER is absorbed into a generic 'error' term, the model conflates pharmacological and psychological sources of improvement. A patient whose 20-point improvement comprises 8 points of BR and 12 points of ER would be indistinguishable from a patient whose 20-point improvement comprises 15 points of BR and 5 points of ER. Yet these two patients have fundamentally different relationships between their biomarker and their drug-specific response. Collapsing BR and ER obscures precisely the signal that biomarker validation requires.

3.4 Time-Varying Response (TV)

Definition. TV is the component of symptom change that occurs independently of any treatment – the natural disease trajectory. It encompasses regression to the mean, the therapeutic effect of regular clinical contact, and the patient's evolving life circumstances.

Clinical example. Even without prazosin, our soldier's PTSD severity is not static. Over the 15 weeks of the trial, he has regular visits with a clinician, keeps a structured schedule, and may experience natural fluctuations in symptom severity related to the calendar (proximity to deployment anniversary), employment, and social support. Some of the 13.8-point improvement seen in the Raskind placebo group reflects TV: structured clinical contact and the passage of time both contribute to symptom change.

Why TV cannot be absorbed into a time covariate. A simple linear time term assumes that symptoms change at a constant rate throughout the trial. In practice, the trajectory is typically nonlinear: rapid early improvement (regression to the mean, engagement with clinical structure) that gradually slows. The Gompertz function captures this S-shaped trajectory. Moreover, TV is individual-specific. Two patients may have very different natural trajectories – one improving steadily, another worsening due to external stressors – and a single time covariate cannot accommodate this heterogeneity.

3.5 Putting the Three Factors Together

Consider our soldier's trajectory through the Hendrickson N-of-1 design (Design 4):

Weeks 1–8 (open-label prazosin):

- BR = 9 points (prazosin accumulates, hyperarousal declines)
- ER = 6 points (full expectancy: he knows he is on drug)
- TV = 4 points (clinical structure, regression to mean)
- Total improvement = 19 points. He feels substantially better.

Weeks 9–12 (blinded discontinuation – randomly assigned to placebo without his knowledge):

- BR = 0 (prazosin washes out within days; no drug, no biological effect)
- ER = 3 points (expectancy at 0.5: he is told he 'may or may not' be on drug)
- TV = 5 points (natural trajectory continues to plateau)
- Total improvement from baseline = 8 points. He notices a setback.

Weeks 13–16 (crossover period 1 – on active drug, blinded):

- BR = 8 points (prazosin restored; biological effect returns)
- ER = 3 points (still blinded; expectancy remains at 0.5)
- TV = 5 points (natural trajectory at plateau)
- Total improvement from baseline = 16 points.

What the N-of-1 design reveals: The contrast between the blinded discontinuation (total = 8) and the crossover period on drug (total = 16) isolates the biological response: 16 - 8 = 8 points, which is the within-person BR estimate. This is far more informative than the group-level 11.3-point difference from the PG-RCT, because it is specific to this patient and uncontaminated by between-person variability in ER and TV.

What a one-component model would conclude: 'Improvement varied from 8 to 19 points across phases – noisy, hard to interpret.'

What the three-component model reveals: 'BR is consistent at 8–9 points when on drug and 0 when off. ER contributes 3–6 points depending on expectancy. TV plateaus at 4–5 points. The drug works for this patient; expectations modulate the total but do not account for the drug-specific effect.'

4 The Biomarker Question

4.1 From Group-Level Efficacy to Individual Prediction

The Raskind et al. (2013) trial established that prazosin is, on average, more effective than placebo for PTSD. But the precision medicine question is different: *Can we predict, before treatment begins, which individual patients will benefit most from prazosin?*

A separate post-hoc analysis by Raskind et al. (2016, *Biol Psychiatry* 80:736–742) found that higher pretreatment standing systolic blood pressure was associated with greater PTSD symptom reduction in soldiers treated with prazosin. This makes mechanistic sense: elevated standing blood pressure may reflect higher baseline CNS adrenergic tone, and patients with more adrenergic dysregulation have more 'room' for prazosin to work.

4.2 Why Biomarker Validation Requires Decomposition

The statistical test that Hendrickson et al. (2020) simulate is not 'Does prazosin work?' but rather 'Does the baseline biomarker (blood pressure) predict the magnitude of the drug-specific response (BR)?'

This distinction matters because the biomarker should predict *only* BR, not ER or TV. If elevated blood pressure predicted expectancy-related improvement, that would be a confound, not a biomarker-treatment

interaction. And if it predicted the time-varying trajectory, the association would be spurious. The three-factor decomposition isolates the target quantity (BR) from the noise (ER + TV), concentrating statistical power on the clinically relevant signal.

The analysis model from Hendrickson et al. (2020) formalizes this as:

$$S_{i,t} = \beta_{i,0} + \beta_1 \cdot bm + \beta_2 \cdot Db + \beta_3 \cdot t + \beta_4 \cdot bm \cdot Db$$

where bm is the baseline biomarker, Db is drug exposure, t is time, and the interaction term β_4 tests whether the biomarker predicts drug-specific response. A non-zero β_4 indicates that the biomarker is predictive – the precision medicine finding.

5 The Modified Gompertz Function

5.1 Why Trajectories Are Not Linear

Each of the three response factors follows a characteristic time course. The improvement does not accumulate at a constant rate; instead, it follows a saturating pattern:

- **Early phase:** Rapid change as the mechanism engages (drug accumulates, expectancy forms, regression to mean operates).
- **Middle phase:** Improvement continues but decelerates.
- **Late phase:** The factor approaches its biological or psychological ceiling.

A linear model would predict indefinite improvement – 2 CAPS points per week, every week, forever. After 50 weeks, that would imply a 100-point improvement on a scale with a maximum of about 136. This is biologically implausible.

The modified Gompertz function captures the saturating pattern with three interpretable parameters:

$$f(t) = \max \cdot e^{-\text{disp} \cdot e^{-\text{rate} \cdot t}}$$

with an offset-rescaling to ensure $f(0) = 0$ and $f(\infty) = \max$.

```
modgompertz <- function(t, maxr, disp, rate) {
  y <- maxr * exp(-disp * exp(-rate * t))
  vert_offset <- maxr * exp(-disp * exp(-rate * 0))
  y <- y - vert_offset
  y <- y * (maxr / (maxr - vert_offset))
  y
}
```

5.2 Parameter Interpretation

Parameter	Symbol	Clinical meaning	Example
Maximum	max	Ceiling effect: the most this factor can contribute	BR max = 11: prazosin can reduce CAPS by at most 11 points
Rate	rate	Speed of onset: how quickly the factor reaches half its ceiling	rate = 0.42: half-maximum BR reached by approximately week 4

Parameter	Symbol	Clinical meaning	Example
Displacement	disp	Curve shape: controls the inflection point and steepness	disp = 5: produces a delayed onset followed by rapid rise

The displacement parameter deserves particular comment. With disp = 5, the Gompertz curve has a pronounced sigmoidal shape: near-zero improvement in the first 1–2 weeks, then a rapid rise, then a plateau. This matches the clinical observation from the Raskind trial that prazosin required a 6-week titration period before reaching therapeutic doses, and that most improvement occurred between weeks 2 and 10.

6 Reconstructing the Raskind Data

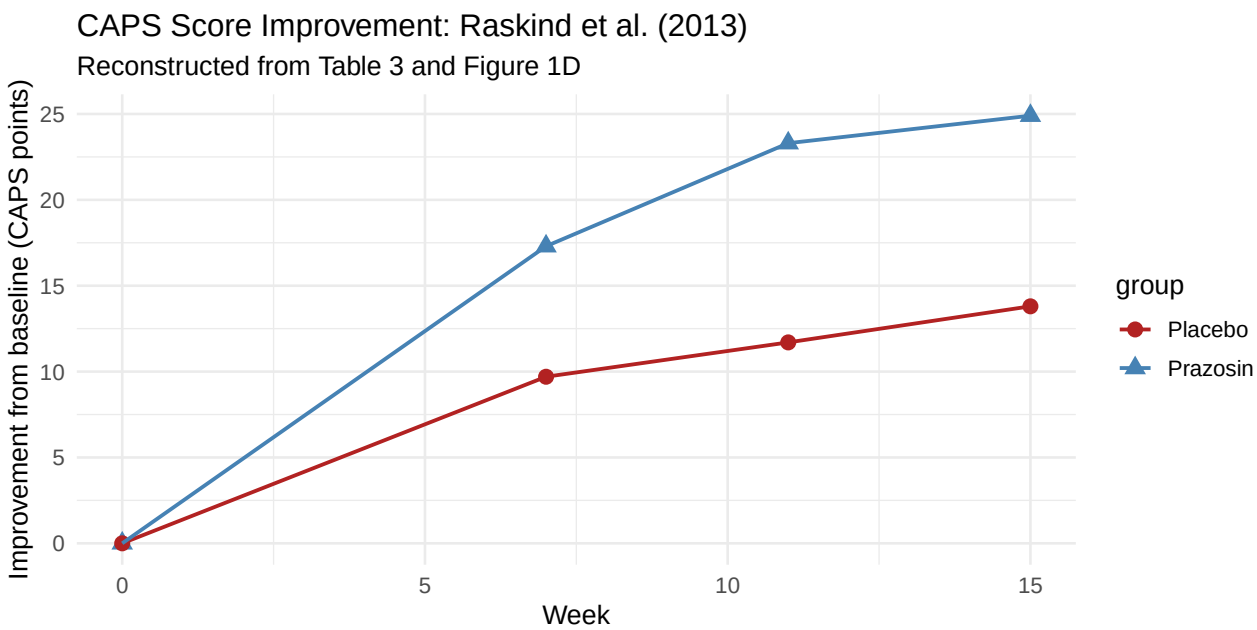
6.1 Extracting CAPS Trajectories

The Raskind et al. (2013) trial assessed CAPS total scores at baseline and weeks 7, 11, and 15. Table 3 of that paper provides adjusted means at baseline and week 15. Weeks 7 and 11 values are interpolated from Figure 1D, which plots the full CAPS total score trajectory for both groups.

```
weeks <- c(0, 7, 11, 15)

caps_prazosin <- c(77.3, 60, 54, 52.4)
caps_placebo  <- c(85.7, 76, 74, 71.9)

trial_data <- data.table(
  week = rep(weeks, 2),
  caps = c(caps_prazosin, caps_placebo),
  group = rep(c("Prazosin", "Placebo"), each = 4)
)

trial_data[, baseline := caps[1], by = group]
trial_data[, improvement := baseline - caps]
```

The baseline CAPS scores differed between groups (prazosin 77.3, placebo 85.7), reflecting the expected variability in a trial of this size (N = 67). The Hendrickson simulation uses the pooled baseline mean of 83.07 (SD = 18.48), averaged across both groups.

```
ggplot(trial_data, aes(x = week, y = improvement,
                       color = group, shape = group)) +
  geom_point(size = 3) +
  geom_line(linewidth = 0.8) +
  labs(
    x = "Week",
    y = "Improvement from baseline (CAPS points)",
    title = "CAPS Score Improvement: Raskind et al. (2013)",
    subtitle = "Reconstructed from Table 3 and Figure 1D"
  ) +
  scale_color_manual(values = c(
    "Placebo" = "firebrick", "Prazosin" = "steelblue"
  )) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "right")
```



Both groups improve over time, but the prazosin group improves more. The gap between the curves widens most between baseline and week 7 (during the titration period), then stabilizes. This widening gap is the observable signature of BR.

6.2 Decomposing the Trajectories into BR and TV+ER

The logic of the decomposition follows directly from the factor definitions:

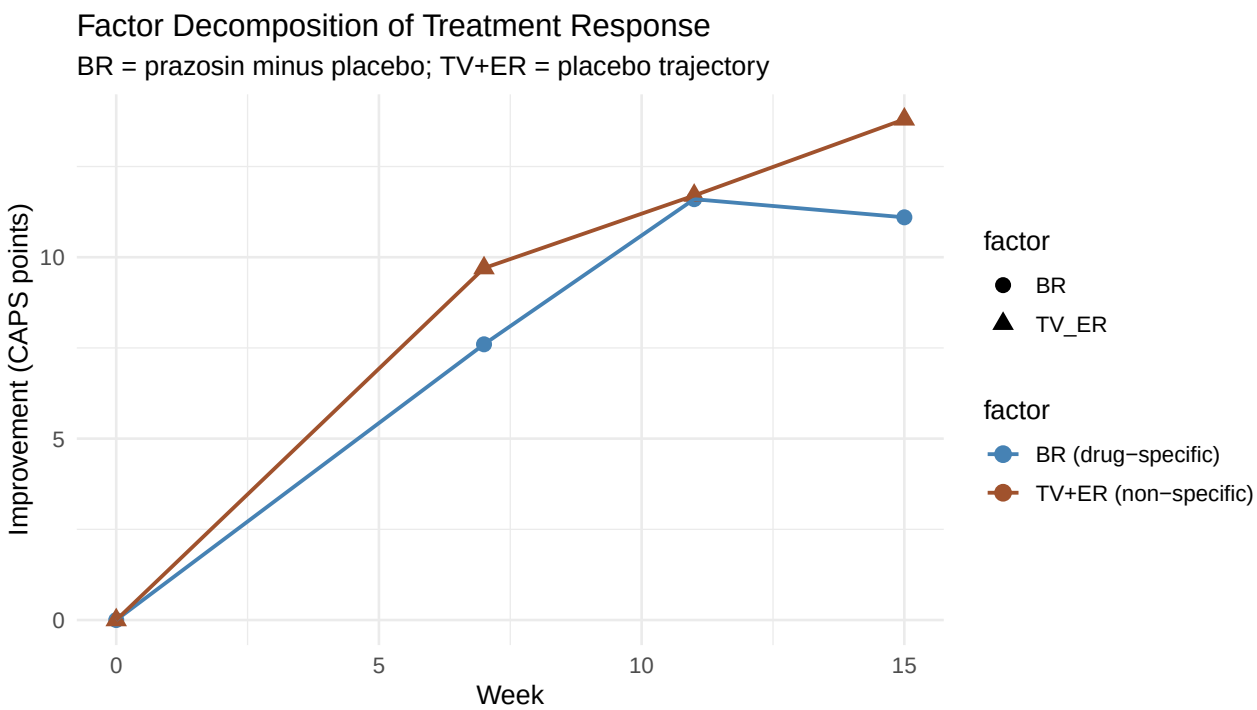
- **Placebo group improvement = TV + ER.** These patients receive no drug, so BR = 0 at all timepoints. Their entire improvement is non-specific: some combination of natural trajectory (TV) and expectancy (ER). Because this was a double-blind trial, placebo participants believed they might be receiving active drug, so ER is non-zero.
- **BR = Prazosin improvement minus Placebo improvement.** Subtracting the placebo trajectory from the prazosin trajectory removes the non-specific components, isolating the drug-specific effect.

```
praz <- trial_data[group == "Prazosin"]
plac <- trial_data[group == "Placebo"]

br_improvement <- praz$improvement - plac$improvement
tver_improvement <- plac$improvement

factor_data <- data.table(
  week = weeks,
  BR = br_improvement,
  TV_ER = tver_improvement
)

knitr::kable(
  factor_data, digits = 2,
  caption = "Factor decomposition of CAPS improvement",
  col.names = c("Week", "BR (drug-specific)",
                "TV+ER (non-specific)")
)
```

Table 3: Factor decomposition of CAPS improvement

Week	BR (drug-specific)	TV+ER (non-specific)
0	0.0	0.0
7	7.6	9.7
11	11.6	11.7
15	11.1	13.8

```r
factor_long <- melt(
  factor_data, id.vars = "week",
  variable.name = "factor",
  value.name = "improvement"
)

ggplot(factor_long, aes(x = week, y = improvement,
                        color = factor, shape = factor)) +
  geom_point(size = 3) +
  geom_line(linewidth = 0.8) +
  labs(
    x = "Week",
    y = "Improvement (CAPS points)",
    title = "Factor Decomposition of Treatment Response",
    subtitle = paste(
      "BR = prazosin minus placebo;",
      "TV+ER = placebo trajectory"
    )
  ) +
  scale_color_manual(
    values = c("BR" = "steelblue", "TV_ER" = "sienna"),
    labels = c("BR (drug-specific)",
               "TV+ER (non-specific)")
  ) +
  theme_minimal(base_size = 12)
```



Several features of this decomposition merit attention. First, the TV+ER component is substantial – comparable in magnitude to BR. By week 15, the placebo group improved by approximately 14 CAPS points, while the drug-specific difference was approximately 11 points. This means that roughly half of the prazosin group's total improvement was non-specific. This is not unusual in psychiatric trials; it is precisely why isolating BR is important for biomarker validation.

Second, the Raskind trial was double-blinded throughout, so the expectancy component is at an intermediate level (participants knew they had a 50% chance of receiving prazosin). In the Hendrickson N-of-1 design, the open-label phase would produce a larger ER contribution, followed by a reduction during blinded phases.

6.3 The 'Tabula Rasa' Assumption

The Raskind data provide a single placebo trajectory that represents TV + ER combined. There is no empirical way to separate TV from ER using these data alone, because every placebo participant simultaneously experienced both the passage of time (TV) and the expectation of possible benefit (ER).

Hendrickson et al. (2020) address this by adopting a 'tabula rasa' (TaRa) parameter set in which TV and ER are assigned identical Gompertz parameters: max = 6.506, disp = 5, rate = 0.35, SD = 10 for each. The combined TV + ER trajectory is simply split equally between the two factors. This is a modeling convenience, not an empirical claim. The sensitivity analyses in Hendrickson Figures 5A and 5B explore what happens when TV and ER parameters diverge.

For the clinician, the practical implication is this: the exact partition of non-specific improvement into TV and ER matters less than the fact that both are separated from BR. The biomarker validation test targets β_4 (the biomarker-by-drug interaction), which depends on BR alone.

7 Fitting the Gompertz Model to the Raskind Data

7.1 Fitting BR (Drug-Specific Response)

We fit the modified Gompertz function to the BR trajectory (prazosin minus placebo improvement at each timepoint) using nonlinear least squares with starting values informed by the Hendrickson publication parameters.

```r
br_fit_data <- data.table(
  t = factor_data$week,
  y = factor_data$BR
)

fit_br <- nlsLM(
  y ~ modgompertz(t, maxr, disp, rate),
  data = br_fit_data,
  start = list(maxr = 12, disp = 5, rate = 0.3),
  lower = c(maxr = 1, disp = 0.1, rate = 0.01),
  upper = c(maxr = 50, disp = 20, rate = 2)
)

br_params <- coef(fit_br)
summary(fit_br)
```

```
##
## Formula: y ~ modgompertz(t, maxr, disp, rate)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## maxr  11.5791     0.8786  13.179   0.0482 *
## disp  20.0000    78.3984   0.255   0.8410
## rate   0.5565     0.5769   0.965   0.5115
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6788 on 1 degrees of freedom
##
## Number of iterations to convergence: 9
## Achieved convergence tolerance: 1.49e-08
```

Fitted BR parameters: max = 11.579, disp = 20.000, rate = 0.557

Hendrickson publication parameters: max = 10.986, disp = 5.000, rate = 0.420

The fitted parameters should approximate the Hendrickson values, since both are derived from the same underlying Raskind data. Any discrepancy reflects the difference between fitting group-level summary statistics (our approach, using 4 timepoints read from published figures) versus fitting individual-level data (the Hendrickson authors had access to the original dataset).

7.2 Fitting TV+ER (Non-Specific Response)

```r
tver_fit_data <- data.table(
  t = factor_data$week,
  y = factor_data$TV_ER
)

fit_tver <- nlsLM(
  y ~ modgompertz(t, maxr, disp, rate),
  data = tver_fit_data,
  start = list(maxr = 15, disp = 5, rate = 0.3),
  lower = c(maxr = 1, disp = 0.1, rate = 0.01),
  upper = c(maxr = 50, disp = 20, rate = 2)
)
```

```r
tver_params <- coef(fit_tver)
summary(fit_tver)
```

```
##
## Formula: y ~ modgompertz(t, maxr, disp, rate)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## maxr  15.6711     6.3557   2.466    0.245
## disp   0.1000     5.4801   0.018    0.988
## rate   0.1375     0.3257   0.422    0.746
##
## Residual standard error: 0.4793 on 1 degrees of freedom
##
## Number of iterations to convergence: 31
## Achieved convergence tolerance: 1.49e-08
```

Fitted TV+ER parameters: max = 15.671, disp = 0.100, rate = 0.137

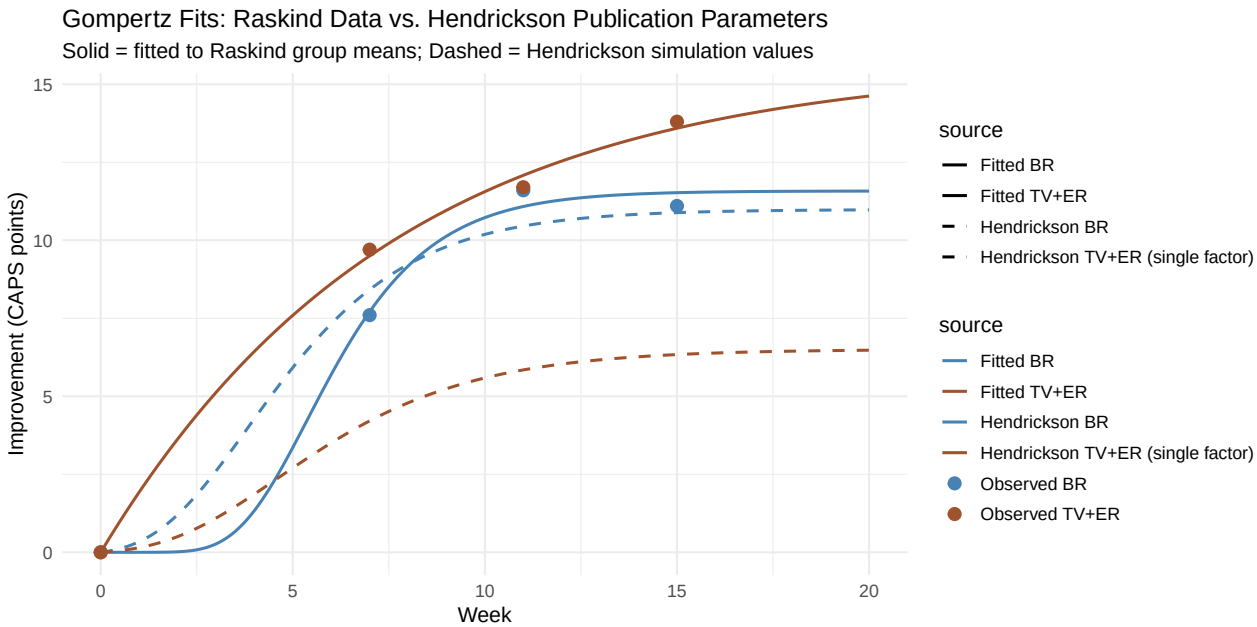
Hendrickson publication parameters (each of TV and ER): max = 6.506, disp = 5.000, rate = 0.350

Note that the Hendrickson max of 6.506 applies to each of TV and ER separately. Under the tabula rasa assumption, TV + ER combined has an effective maximum of approximately $2 \times 6.506 = 13.0$ CAPS points. The fitted TV+ER maximum from the Raskind placebo data should approximate this value.

7.3 Comparing Fitted and Publication Curves

```r
t_smooth <- seq(0, 20, by = 0.1)

br_curve <- data.table(
  t = t_smooth,
  y = modgompertz(t_smooth, br_params["maxr"],
                  br_params["disp"], br_params["rate"]),
  source = "Fitted BR"
)
tver_curve <- data.table(
  t = t_smooth,
  y = modgompertz(t_smooth, tver_params["maxr"],
                  tver_params["disp"], tver_params["rate"]),
  source = "Fitted TV+ER"
)

br_hend <- data.table(
  t = t_smooth,
  y = modgompertz(t_smooth, 10.986, 5, 0.42),
  source = "Hendrickson BR"
)
tver_hend <- data.table(
  t = t_smooth,
  y = modgompertz(t_smooth, 6.506, 5, 0.35),
  source = "Hendrickson TV+ER (single factor)"
)

curves <- rbind(br_curve, tver_curve, br_hend, tver_hend)
```

```r
obs <- rbind(
  data.table(t = weeks, y = factor_data$BR,
             source = "Observed BR"),
  data.table(t = weeks, y = factor_data$TV_ER,
             source = "Observed TV+ER")
)

ggplot() +
  geom_line(
    data = curves,
    aes(x = t, y = y, color = source, linetype = source),
    linewidth = 0.8
  ) +
  geom_point(
    data = obs,
    aes(x = t, y = y, color = source),
    size = 3
  ) +
  scale_color_manual(values = c(
    "Fitted BR" = "steelblue",
    "Hendrickson BR" = "steelblue",
    "Fitted TV+ER" = "sienna",
    "Hendrickson TV+ER (single factor)" = "sienna",
    "Observed BR" = "steelblue",
    "Observed TV+ER" = "sienna"
  )) +
  scale_linetype_manual(values = c(
    "Fitted BR" = "solid",
    "Hendrickson BR" = "dashed",
    "Fitted TV+ER" = "solid",
    "Hendrickson TV+ER (single factor)" = "dashed"
  )) +
  labs(
    x = "Week",
    y = "Improvement (CAPS points)",
    title = paste(
      "Gompertz Fits: Raskind Data vs.",
      "Hendrickson Publication Parameters"
    ),
    subtitle = paste(
      "Solid = fitted to Raskind group means;",
      "Dashed = Hendrickson simulation values"
    )
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "right")
```



The dashed Hendrickson TV+ER curve represents a *single* factor (either TV or ER alone, each with max = 6.506). The solid fitted TV+ER curve represents the *combined* non-specific trajectory from the placebo group. The Hendrickson curve should therefore lie below the fitted curve, roughly at half its height.

8 Variance and the Covariance Structure

8.1 Between-Person Variability in Each Factor

The Gompertz parameters describe the *mean* trajectory. But patients vary: one soldier's BR might be 12 CAPS points while another's is only 4. This between-person variability is captured by the standard deviation of each factor, which enters the simulation through the covariance matrix used to generate individual trajectories.

From Raskind Table 3, we can recover the raw CAPS total score SD from the 95% confidence intervals:

- Prazosin: mean = 77.3, 95% CI = 69.1–85.5, giving SE = 4.1 and SD = $4.1 \times \sqrt{32} = 23.2$
- Placebo: mean = 85.7, 95% CI = 78.0–93.3, giving SE = 3.9 and SD = $3.9 \times \sqrt{35} = 23.1$

For the change-from-baseline SD, the reported difference in change is 11.3 points (95% CI = 2.0–20.7). The SE of the difference is $(20.7 - 2.0)/(2 \times 1.96) = 4.77$. Assuming equal SDs across groups:

$$4.77 = \text{SD} \times \sqrt{1/32 + 1/35} = \text{SD} \times 0.247$$

This yields an SD of change of approximately 19.3 CAPS points.

The Hendrickson factor SDs are substantially smaller than this total SD because they represent the variability of each individual factor, not the total outcome. The total outcome variability is the sum of variances across all factors plus baseline variability, partially offset by cross-factor correlations.

Hendrickson factor standard deviations:

Factor	SD
BR (drug-specific)	8
TV (time-variant)	10
ER (expectancy)	10

Factor	SD
Baseline (BL)	18.5

```
total_var <- 18.5^2 + 10^2 + 10^2 + 8^2
```

The sum of marginal variances is 606, yielding an implied total SD of 24.6. The observed CAPS SD is approximately 23, which is lower than this implied value. This is expected: the factors are not independent, and cross-correlations (particularly between baseline severity and the response factors) reduce the total variance below the sum of its components.

8.2 Why Separate Variances Matter

Each factor has its own variance because the underlying mechanisms operate on different timescales and are subject to different sources of individual variation:

- **BR variance** (SD = 8) reflects individual differences in pharmacokinetics: metabolism rate, receptor density, drug absorption. Two soldiers on identical prazosin doses may achieve different CNS drug concentrations.
- **TV variance** (SD = 10) reflects heterogeneity in natural disease trajectory: life circumstances, comorbidities, social support, proximity to trauma cues.
- **ER variance** (SD = 10) reflects individual differences in suggestibility, prior treatment experience, and the strength of the therapeutic alliance.

The simulation generates correlated factor values from a multivariate normal distribution with a covariance matrix that encodes both the marginal variances and the between-factor correlations (including the correlation between baseline biomarker and BR, which is the target of the validation test).

9 Parameter Comparison

```
comparison <- data.table(
  Factor = c("BR", "TV+ER (combined)",
             "TV (Hendrickson)", "ER (Hendrickson)"),
  max_fitted = c(br_params["maxr"], tver_params["maxr"],
                 NA, NA),
  max_hendrickson = c(10.986, NA, 6.506, 6.506),
  disp_fitted = c(br_params["disp"], tver_params["disp"],
                  NA, NA),
  disp_hendrickson = c(5, NA, 5, 5),
  rate_fitted = c(br_params["rate"], tver_params["rate"],
                  NA, NA),
  rate_hendrickson = c(0.42, NA, 0.35, 0.35),
  sd_hendrickson = c(8, NA, 10, 10)
)

knitr::kable(
  comparison, digits = 3,
  caption = paste(
    "Comparison of parameters fitted to Raskind",
    "group means vs. Hendrickson publication values"
  )
)
```

Table 5: Comparison of parameters fitted to Raskind group means vs. Hendrickson publication values

Factor	max_fitted	max_hendrickson	disp_fitted	disp_hendrickson	rate_fitted	rate_hendrickson	sd_hendrickson
BR	11.579	10.986	20.0	5	0.557	0.42	8
TV+ER (combined)	15.671	NA	0.1	NA	0.137	NA	NA
TV (Hendrickson)	NA	6.506	NA	5	NA	0.35	10
ER (Hendrickson)	NA	6.506	NA	5	NA	0.35	10

10 How Trial Design Affects Power: Lessons from the Simulation

10.1 The Four Designs Compared

Hendrickson et al. (2020) compared four trial architectures, all matched in total duration and number of assessment points (see their Figure 2):

1. **Open-label (OL):** 20 weeks of open-label prazosin. All participants know they are on drug. BR and ER are fully confounded; the biomarker-by-drug interaction cannot be tested directly.
2. **Open-label + blinded discontinuation (OL+BDC):** 16 weeks of open-label prazosin, then a 4-week blinded discontinuation. The discontinuation block introduces a within-person contrast but is brief.
3. **Traditional crossover (CO):** Two 10-week periods, randomized to drug-then-placebo or placebo-then-drug. Both periods are blinded. Provides within-person contrast but no open-label stabilization phase.
4. **Proposed N-of-1 design:** 8 weeks open-label, 4 weeks blinded discontinuation, then two 4-week crossover blocks. This design combines the clinical advantages of open-label stabilization with the statistical advantages of multiple within-person contrasts.

10.2 Key Findings

The simulation results (Hendrickson Figure 4A) demonstrate that:

- **Design 4 (N-of-1) had the highest power** for detecting the biomarker-treatment interaction under the tabula rasa parameter set, outperforming both the open-label and OL+BDC designs. With 70 subjects and a biomarker-BR correlation of 0.6, the N-of-1 design achieved power of approximately 0.99 without censoring.
- **The traditional crossover was comparable** in power to the N-of-1 design (also approximately 0.99 with 70 subjects), but lacked the clinical advantage of beginning with open-label stabilization.
- **The open-label design performed poorly** (power of approximately 0.25 with 70 subjects), because it cannot separate BR from ER.
- **Carryover was devastating to power** (Hendrickson Figure 4B). Even a short carryover half-life of 0.1 weeks reduced N-of-1 power substantially, because it blurs the on-drug/off-drug contrast that the design depends on. For prazosin, with its 6–10 hour duration of action, the pharmacokinetic half-life is approximately 0.02–0.03 weeks, so carryover should be negligible between weekly assessments.

10.3 Clinical Implications

For the clinician considering an N-of-1 trial to validate a predictive biomarker, the Hendrickson results suggest several principles:

- **Include an open-label phase** to stabilize acutely symptomatic patients and maximize enrollment. This is ethically important: patients referred for PTSD treatment are unlikely to accept initial randomization to placebo.
- **Choose drugs with short half-lives** (or at least short pharmacodynamic durations). Carryover effects erode the power advantage of crossover designs. Prazosin, with its 6–10 hour duration of action, is well suited to this approach.
- **Maximize the number of within-person contrasts.** The N-of-1 design's advantage comes from having multiple on-drug and off-drug periods within the same patient, which averages out the noise from any single transition.
- **The signal-to-noise ratio matters more than sample size.** Figure 5A of Hendrickson et al. shows that when the BR maximum is large relative to the TR and ER maximums, power approaches 1.0 even with modest sample sizes. The practical implication is that drugs with strong biological effects and short durations of action are ideal candidates for N-of-1 biomarker validation.

11 Sensitivity to Gompertz Parameters: What Figure 5 Tells Us

The preceding sections establish that the simulation's power depends on the trial design and the biomarker-BR correlation. But a question of equal practical importance is: *How sensitive are these power estimates to the Gompertz parameters themselves?* If the true trajectory parameters differ from the tabula rasa values, do the design recommendations still hold?

Hendrickson et al. (2020) address this directly in their Figure 5, which systematically varies the maximum response values (Figure 5A) and the rate parameters (Figure 5B) across all three factors and all four trial designs. These results have important implications for clinicians considering which drugs and which outcome measures are amenable to N-of-1 biomarker validation.

11.1 Effect of Maximum Response Values (Figure 5A)

Figure 5A presents a 3 x 3 grid of heat maps for each trial design. The rows correspond to the BR maximum (5, 8, or 11 CAPS points), and the columns correspond to the ER maximum (5, 8, or 11), with each cell within a panel varying the TR maximum (5, 8, or 11). The biomarker-BR correlation is fixed at 0.6 and N = 35.

The results reveal a clear and clinically interpretable pattern:

Increasing BR max improves power across all designs. This is the signal: a larger drug-specific ceiling means more separation between high-biomarker and low-biomarker patients, making the interaction easier to detect. When BR max = 11 (roughly equivalent to the tabula rasa value of 10.986), power exceeds 0.90 for the crossover and N-of-1 designs regardless of the ER and TR maximums. When BR max drops to 5 (a weak drug effect), power falls below 0.20 for the open-label design and remains marginal even for the crossover.

Clinical translation: A drug that produces at most a 5-point improvement on the CAPS (out of a maximum possible score of approximately 136) is a poor candidate for biomarker validation via any design. The drug effect is simply too small relative to the noise floor. Prazosin, with its estimated BR max of approximately 11 CAPS points, sits in a favorable region of this parameter space.

Increasing ER max decreases power, especially for designs with expectancy variation. Larger non-specific expectancy effects add variance to the outcome without contributing to the biomarker-BR signal. This dilution is most pronounced in the OL+BDC and N-of-1 designs, which include open-label phases

where expectancy is at its maximum (1.0) followed by blinded phases where it drops to 0.5. The resulting within-person ER change adds noise to the treatment contrast.

Clinical translation: Conditions with large placebo responses (e.g., depression, chronic pain) will require larger sample sizes or stronger drug effects to achieve adequate power. Conditions where the placebo response is modest (e.g., neurodegenerative diseases with objective biomarkers) may be more favorable for N-of-1 designs.

Increasing TR max also decreases power. The time-varying component is pure noise from the perspective of biomarker validation. It affects all participants equally regardless of drug assignment, so it does not bias the interaction estimate, but it inflates the residual variance and reduces the signal-to-noise ratio.

Clinical translation: Diseases with highly variable natural trajectories will require larger samples. One strategy to mitigate this is to shorten the trial (reducing the window over which TV accumulates), but this must be balanced against the need for sufficient treatment exposure to observe BR.

The ratio of BR max to non-specific maximums is the key quantity. When BR max = 11 and both ER max and TR max = 5 (a favorable signal-to-noise ratio), even the open-label design achieves power above 0.60. When the ratio reverses (BR max = 5, ER max = TR max = 11), power is uniformly poor. This highlights that the choice of drug and outcome measure matters as much as the choice of trial design.

11.2 Effect of Rate Parameters (Figure 5B)

Figure 5B uses the same grid layout but varies the rate parameters (0.2, 0.35, or 0.5 per week) for each factor, holding the maximums at their tabula rasa values. The biomarker-BR correlation is fixed at 0.3 (lower than in Figure 5A), making the power landscape more conservative.

The rate parameter controls how quickly each factor approaches its ceiling. A rate of 0.5 means the factor reaches half-maximum in approximately 2–3 weeks; a rate of 0.2 means this takes 5–7 weeks. The results are less straightforward than for the maximums:

For designs with blinded discontinuation (OL+BDC and N-of-1), faster BR rates generally improve power. A higher BR rate means the drug effect ramps up quickly during on-drug periods and (because the offset-rescaling enforces $f(0) = 0$) disappears quickly when the drug is removed. This sharpens the within-person contrast between on-drug and off-drug assessments.

For the traditional crossover, faster BR rates paradoxically *decrease* power across much of the parameter space. This occurs because in the crossover design, each treatment period is 10 weeks long. When BR reaches its ceiling quickly (say by week 3), the remaining 7 weeks of the period provide assessment points that are all at the same plateau value. These 'ceiling' observations contain no additional information about the rate of onset but do accumulate TV and ER variance. In effect, the crossover design wastes assessment points when the drug effect saturates early.

Clinical translation: The relationship between drug onset speed and trial design is not monotonic. For drugs with very fast onset (analogous to high BR rate), the N-of-1 design with its shorter treatment blocks (4 weeks) is more efficient than the crossover with its longer blocks (10 weeks). For drugs with slower onset, longer treatment blocks may be necessary to observe the full BR trajectory, favoring the crossover design.

Faster ER rates decrease power for the OL+BDC and N-of-1 designs. When the expectancy response ramps up quickly, it reaches its ceiling during the open-label phase and is fully established before the blinded transition. This means the ER drop upon entering the blinded phase is large and abrupt, adding a substantial within-person change that is unrelated to BR. The analysis model must absorb this ER-driven variance, reducing the precision of the biomarker-by-drug interaction estimate.

TR rate has minimal impact on power across most of the parameter space and most designs. This is because the time-varying component is continuous and gradual: whether it reaches its plateau by week 5 or week 10 has little effect on the within-person contrasts that drive power in the crossover and N-of-1 designs.

11.3 Practical Synthesis

The Figure 5 results can be distilled into a set of practical guidelines for clinicians and trialists:

1. **Screen candidate drugs by their signal-to-noise profile.** Before committing to an N-of-1 design, estimate the likely BR max from prior RCT data (as we did from the Raskind trial) and compare it to the expected ER and TR maximums. If BR max is less than half of ER max + TR max, power will be poor regardless of design.
2. **Match block length to drug onset speed.** Fast-onset drugs (high BR rate) benefit from shorter treatment blocks with more transitions (the N-of-1 design). Slow-onset drugs may require longer blocks to observe the full BR trajectory (favoring the crossover).
3. **Account for the placebo response in power calculations.** Conditions with large expectancy effects require larger sample sizes. The tabula rasa assumption (ER max = TR max = 6.5) may be optimistic for conditions like depression, where placebo response rates often exceed 30%.
4. **The Gompertz parameters are not nuisance quantities.** They directly affect the feasibility of biomarker validation. Reporting and estimating these parameters from pilot data should be a standard part of N-of-1 trial planning.

12 Limitations of This Reconstruction

Several caveats apply to the Gompertz parameter estimates derived here:

- **Only four timepoints** (weeks 0, 7, 11, 15) are available from the Raskind publication, providing minimal constraint on a three-parameter model. With four data points and three parameters, there is only one residual degree of freedom.
- **Weeks 7 and 11 values were read from Figure 1D** of Raskind et al. (2013) and are approximate. The published table provides exact values only at baseline and week 15.
- **We are fitting group means, not individual data.** The Hendrickson authors had access to the original individual-level data from the Raskind trial, which would provide far more constraint on the Gompertz fit. Our reconstruction is a pedagogical exercise, not a replication.
- **The displacement parameter is poorly constrained.** With only four timepoints, the data cannot distinguish between different values of disp that produce similar curves over the observed time window. The Hendrickson choice of disp = 5 for all factors was likely guided by the individual-level data and by theoretical considerations about the shape of treatment response curves.

13 Summary

The Hendrickson et al. (2020) simulation makes a compelling case for aggregated N-of-1 designs in predictive biomarker validation. The key conceptual ingredients are:

1. **Decomposition of treatment response** into biological (BR), expectancy (ER), and time-varying (TV) components, each following a Gompertz trajectory with parameters derived from the Raskind et al. (2013) prazosin trial.
2. **Isolation of the target signal** (the biomarker-by-BR interaction) from the noise (ER + TV), which concentrates statistical power on the precision medicine question.
3. **Within-person contrasts** achieved through blinded discontinuation and crossover phases, which provide far more efficient estimation of individual-level BR than between-group comparisons.
4. **Practical clinical design** that begins with open-label stabilization to address ethical and enrollment concerns, while preserving statistical power through subsequent blinded phases.

The Gompertz parameters are not arbitrary simulation inputs. They are grounded in observable clinical data: the shape and magnitude of CAPS score trajectories in a real trial of prazosin for combat-related PTSD. Understanding where these parameters come from – and what they represent clinically – is essential for interpreting the simulation results and for designing future N-of-1 trials in precision psychiatry.

14 References

- Hendrickson RC, Thomas RG, Schork NJ, Raskind MA (2020). Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Front Digit Health* 2:13.
- Raskind MA, Peterson K, Williams T, et al. (2013). A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *Am J Psychiatry* 170:1003–1010.
- Raskind MA, Peskind ER, Chow B, et al. (2016). Trial of prazosin for post-traumatic stress disorder in military veterans. *Biol Psychiatry* 80:736–742.

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